

PROBLEMS ON POLYMERS

1. The molecular weight of PMMA is 250 000 g/mol. If all the chains have the same size, calculate (a) the degree of polymerization and (b) number of chains in 1 g of polymer.

Data: Atomic mass (g/mol): O = 16, H = 1, C = 12.

(Solution: (a) 2500; (b) 2.408×10^{18} chains)

2. Describe for each of the following polymer pairs, which one is more likely to crystallize? Explain your answer.

- a) Branched Polyethylene vs linear polyethylene.
- b) Polyethylene vs polyethylene-polypropylene co-polymer.
- c) Isotactic Polypropylene vs Atactic polypropylene.

(Solution: a) Linear PE, b) PE, c) Isotactic PP)

3. Assuming that a thermoplastic polymer can be produced by rolling (deformation) and casting (with fast cooling). Which one of the two methods will produce a polymer with higher strength? Explain your answer.

(Solution: polymer produced by rolling)

4. Which of the following polymer pairs has higher melting temperature? Why?

- a) Branched PE with $\langle M_n \rangle = 850\,000$ g/mol; Linear PE with $\langle M_n \rangle = 850\,000$ g/mol
- b) PTFE with density equal to 2.14 g/cm³ and $\langle M_w \rangle = 600\,000$ g/mol; PTFE with density equal to 2.20 g/cm³ and $\langle M_w \rangle = 600\,000$ g/mol.
- c) Syndiotactic PVC with $\langle M_n \rangle = 500\,000$ g/mol; Linear PE with $\langle M_n \rangle = 225\,000$ g/mol.
- d) Linear and syndiotactic PP with $\langle M_w \rangle = 750\,000$ g/mol; Atactic and linear PP with $\langle M_w \rangle = 500\,000$ g/mol.

(Solution: a) Linear PE > Branched PE; b) PTFE 2.20 > PTFE 2.14; c) PVC > PE; d) Syndiotactic PP > Atactic PP)

5. Draw three representations of the logarithm of the Young's modulus *versus* temperature for an amorphous polymer.

- a) In one of those show how the behavior changes when the molecular weight increases.
- b) In another diagram, show the behavior changes when crosslinking increases.
- c) If this polymer is processed so that it has some crystallinity, show how the Young's modulus changes with temperature.

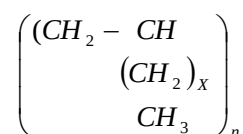
6. Determine the degree of crystallinity of polypropylene with density 0.853 g/cm³, given that the densities of amorphous and totally crystalline polypropylene are 0.83 and 0.947 g/cm³ respectively. Represent the change in the elastic modulus with temperature for this polymer, a completely amorphous polymer and a completely crystalline polymer.

(Solution: 21.8%)

7. A thermoplastic polymer has been analyzed, obtaining the following fractions of weight w_i shown in the table for the following ranges of molecular weight in the table:

- Calculate the weight-average molecular weight
- Calculate the polydispersity assuming that the number average molecular weight is 12880 g/mol.
- If the polymerization degree is 230 and assuming that the polymer is an alkene (C_nH_{2n}) with the general formula shown below, determine the value of X in the structural formula below. (Atomic mass: H = 1, C = 12).
- Explain if the crystallinity of the polymer should be higher or lower than the crystallinity of polyethylene.

Molecular weight (g/mol)	w_i
0-5000	0.02
5000-10000	0.12
10000-15000	0.32
15000-20000	0.32
20000-25000	0.22



(Solution: a) $M_w = 15500$ g/mol; b) $PDI = 1.2$; c) $X = 1$, polybutadiene; d) lower)

8. A polymer sample with unknown molecular weight has been analyzed and the following results were obtained.

- Calculate the number-average and weight-average molecular weight.
- If the number average degree of polymerization is 211, calculate the molecular weight of the monomeric unit.
- Which of these polymers correspond to: PE, PVC, PTFE, PP, PS, PMMA, Nylon 66, PC.
- What is the weight-average degree of polymerization?

Molecular weight range (g/mol)	Number Fraction, x_i	Weight Fraction w_i
8000-20000	0.05	0.02
20000-32000	0.15	0.08
32000-44000	0.21	0.17
44000-56000	0.28	0.29
56000-68000	0.18	0.23
68000-80000	0.10	0.16
80000-92000	0.03	0.05

(Solution: a) $M_n = 47720$, $M_w = 53720$; b) $M_0 = 226.2$; c) Nylon 66; d) $X_w = 238$)

9.- The densities of amorphous and totally crystalline polypropylene are given in the table below.

- Write the formula of polypropylene and justify whether it can present tacticity.
- Calculate the density of completely crystalline and completely amorphous polypropylene.
- Determine the density of a sample that is 74.6% crystalline.

ρ (g/cm ³)	Crystallinity (%)
0.904	62.8
0.895	54.4

(Solution: b) $\rho_a = 0.84$ g/cm³, $\rho_c = 0.946$ g/cm³; c) $\rho = 0.917$ g/cm³)